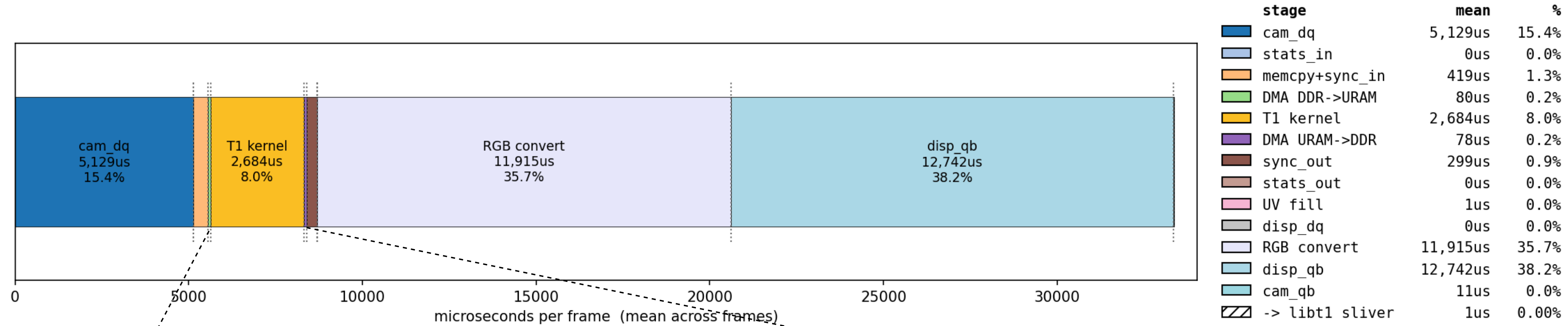
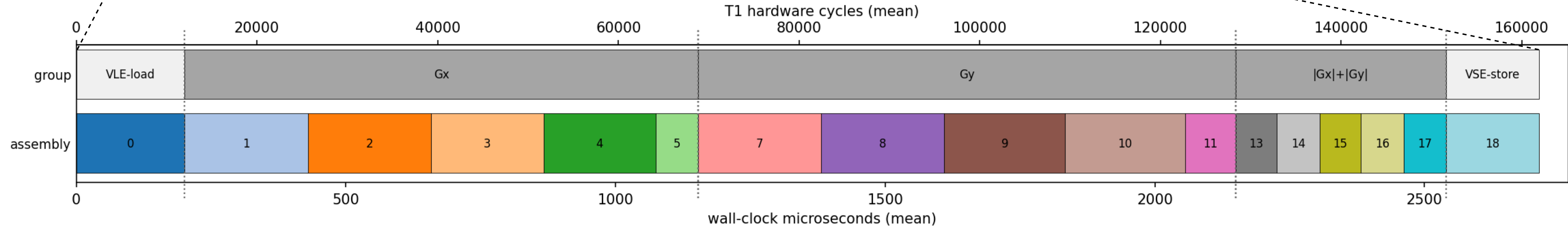


VisionSoC pipeline per-frame breakdown N=10000 frames (skipping frame 0 warm-up)
mean frame = 33,357us ~30.0 fps T1 kernel = 2,684us wall = 2,683us T1@60MHz + 1us libt1/Linux



Sobel kernel: per-instruction breakdown 19 issued ops + 2 dispatcher-only CSR flips
total T1 cycles = 161.9k total wall = 2,712us libt1 sliver = 14us (@60MHz)



stage assembly	group	cycles	wall
0: vle8.v	VLE-load	12.0k	201us
1: vmv.v.i	Gx pass	13.7k	229us
2: vmv.v.i	Gx pass	13.6k	228us
3: vslidedown.vi	Gx pass	12.5k	209us
4: vslideup.vi	Gx pass	12.4k	207us
5: vsub.vv	Gx pass	4.7k	79us

stage assembly	group	cycles	wall
6: csrwi(vmode=1) CSR flip	CSR flip	0	0us
7: vmv.v.i	Gy pass	13.6k	228us
8: vmv.v.i	Gy pass	13.6k	228us
9: vslidedown.vi	Gy pass	13.4k	224us
10: vslideup.vi	Gy pass	13.3k	222us
11: vsub.vv	Gy pass	5.6k	93us

stage assembly	group	cycles	wall
12: csrwi(vmode=0) CSR flip	CSR flip	0	0us
13: vrsb.vi	Gx + Gy	4.5k	77us
14: vmax.vv	Gx + Gy	4.7k	79us
15: vrsb.vi	Gx + Gy	4.5k	76us
16: vmax.vv	Gx + Gy	4.7k	79us
17: vsadd.vv	Gx + Gy	4.7k	79us
18: vse8.v	VSE-store	10.3k	172us